Certified H_∞ co-design of actuator, sensor and controller for the linearised Ginzburg–Landau equation

Version 7 — frozen campaign report

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Abstract

We formulate and solve a simultaneous ("co-design") optimisation of the actuator distribution, the sensor distribution and a fixed-order dynamic compensator, so as to minimise the closed-loop H_∞ norm of the energy resolvent of the linearised complex Ginzburg–Landau (GL) equation in its subcritical, globally stable regime. The study is a methodological stepping stone towards flow control of a two-dimensional open cavity at $Re = 7500$: every algorithmic ingredient is chosen so that it survives the transition to a large-scale discretisation, and the GL benchmark is used to validate it against reference solutions that remain computable by dense linear algebra.

This revision freezes the campaign. Relative to version 6 it adds: (i) a state-of-the-art review positioning the method against the large-scale H_∞-computation and nonsmooth-synthesis literature; (ii) an *unconditional* stability certificate based on the argument principle, exploiting the sparse-plus-low-rank structure of the closed loop, cross-validated against $104/104$ dense eigenvalue counts; (iii) a measured resolution of the trust-region question ($20\times$ fewer oracle calls at equal or better γ); (iv) a complete account of the mirror symmetry of the problem, including a certified instance of *weak spontaneous symmetry breaking* at the fixed-compensator optimum, healed by a single free complex gain; (v) substantially improved optima — the headline case improves from $\gamma = 4.19$ (v5) and 3.93 (v6) to a certified $\gamma = 3.9255$, and the gain-plus-position case K+B+C from 8.03 to 7.96 — every one of the 27+3 reported designs now carrying four independent certificates; and (vi) an honest audit of the pitfalls encountered, several of which were found by *distrusting figures*, together with a quantified specification of the inner-solver replacement that the next campaign requires. The uncontrolled amplification $\gamma_{\mathrm{ol}} = 843.2$ is reduced by a factor of 86 by position-and-gain optimisation and by a factor of 215 when the actuator and sensor are optimised as free fields, within 12% of the provable two-channel floor $\sup_\omega \sigma_2 = 3.509$.

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1 Motivation

Classical flow-control practice separates two decisions that are coupled: hardware placement (guided by open-loop sensitivity or resolvent analysis) and compensator synthesis for the resulting plant. The achievable closed-loop performance is a property of the *triple* (actuator, sensor, compensator), and the placement that is optimal in an open-loop sense need not be optimal once feedback is closed. Following Chen & Rowley's H_2 study, the present programme (unchanged since v6, and only summarised here) optimises the worst-case energy amplification — the H_∞ norm of the energy resolvent — over (i) actuator/sensor *positions* (Gaussians of fixed width), (ii) actuator/sensor *fields* (arbitrary complex distributions of unit L^2 norm with Sobolev regularisation), and (iii) a fixed-order ($r = 8$, tridiagonal) compensator, in all $2^{\cdot}$ combinations forming a **27-case nested suite** $\{\text{none}, \text{K}, \text{KSP}, \text{KGC}\} \times \{-, \text{B}, \text{C}, \text{B+C}, \text{FB}, \text{FC}, \text{FB+FC}\}$ whose partial order ($\gamma_{\text{child}} \leq \gamma_{\text{parent}}$) is used throughout as a free consistency check. Production parameters: $n_e = 200$ elements ($n = 399$ complex dofs), stability margin $\alpha \leq -\varepsilon$ with $\varepsilon = 0.05$, certified frequency band $\omega \in [-30, 30]$, compensator pole cap $|\operatorname{Im} \lambda(A_c)| \leq 20$.

The requirement inherited from the cavity is uncompromising: every ingredient must be matrix-free or sparse-direct, parallelisable, and *certified* — a locally convergent H_∞ evaluation inside the optimisation loop silently tracks a local maximiser of $\sigma_1(\omega)$ and the optimiser then minimises the wrong functional (v6, §3.5). This report documents the frozen state of the resulting pipeline and, equally importantly, of its validation apparatus.

2 Relation to the state of the art

This section, new in v7, positions each block of the pipeline against the literature and records why the present choices were made. Two bodies of work are relevant: large-scale H_∞/L_∞-norm *computation*, and nonsmooth *synthesis* of structured controllers.

2.1 Computing the H_∞ norm at scale

The classical algorithm of Boyd–Balakrishnan and Bruinsma–Steinbuch [1] is a level-set iteration on the imaginary-axis eigenvalues of a $2n \times 2n$ Hamiltonian: globally convergent, quadratic, and certified — but strictly dense, $O(n^3)$ per level. It is the oracle inside `hinfstruct`/`systune`, and the survey [12] concedes it is the scaling bottleneck beyond a few hundred states. The sparse alternatives iterate on spectral value sets [2] or structured pseudospectra of descriptor pencils [3], using only sparse solves — but both are *locally* convergent and state so: they return lower bounds with no globality guarantee. Two later works resolve the tension. Benner & Mitchell [5] rebalance the workload: cheap local Newton optimisation of $\sigma_1(\omega)$ over frequency (each evaluation a sparse solve; the second derivative is Lipschitz at maximisers, licensing Newton) supplies high-quality peaks, and the expensive global level-set test is invoked only to certify or to jump between local maximisers; they also note the method is embarrassingly parallel over intervals and multistarts. Schwerdtner, Mengi & Voigt [6] certify a candidate level γ *globally* by adaptive interval elimination with structure-preserving shift-invert Arnoldi on the associated even pencil — eigenvalue-free "disks" whose union exhausts the frequency axis.

The oracle used here (v6 §3.6, unchanged) independently reproduces the [5] architecture — grid seeds, local Newton peak polishing, warm-started weighted power iterations for the operator norm of the full-field channel — with the crucial substitution of [6]-style interval elimination for the dense pencil, which is the only certified test that survives at cavity scale. One structural point favours the present setting: [2] and the greedy interpolatory subspace method of [4] assume few inputs and outputs, whereas the performance channel here is *full-field* ($d \to z$ with B_1, C_1 of full rank); the operator-norm power iteration applies where their rank-one machinery does not. The greedy-subspace idea nevertheless remains the most promising *accelerator* for the cavity (§8): Hermite-interpolatory reduced models make the whole gain curve cheap after a few full-order solves, with superlinear convergence in subspace

dimension.

2.2 Nonsmooth synthesis: bundle versus exchange

The `hinfstruct` lineage [7, 8] treats $\gamma(\theta) = \max_\omega \sigma_1(\omega;\theta)$ as a nonsmooth function of the design θ: descent directions are computed from the Clarke subdifferential enriched by *extended sets of frequencies* (near-active peaks), and `systune` [9, 12] wraps this in a proximity-control *bundle* method — cutting planes, aggregation, and a proximity parameter τ managed like a trust radius — with a convergence theory for lower-C^1 functions, plus a progress function $\Phi_\eta = \max\{f, \eta g\}$ with bisection on η for multi-objective designs. GRANSO's BFGS-SQP [13] attacks the same nonsmooth max directly with a steering exact penalty; its authors state plainly that it carries no convergence guarantees. Closest to the present physical setting, Apkarian & Noll's infinite-dimensional paper [11] optimises on frequency grids with a-posteriori re-certification and — decisive for us, see §4.3 — replaces spectral-abscissa barriers, which they note are unreliable because "values redescend" after a crossing, by a *Nyquist winding-number* stability certificate with first-order grid-refinement bounds.

The method used here is the *exchange* dialect of the same "finitely many active frequencies" structure: a finite set Ω of frequency constraints $\sigma_1(\omega_i;\theta) \le t$ is maintained explicitly (the exchange set *is* the bundle, kept in frequency space), the restricted epigraph problem is solved to local optimality, and newly located peaks are exchanged in. Two consequences justify the choice. First, with Ω fixed, each constraint is *smooth* in θ wherever $\sigma_1(\omega_i)$ is simple: the exchange dissolves precisely the nonsmoothness the bundle machinery exists to manage, which is why the BFGS-SQP failures documented in v6 (§3.5: stalls at 6–11 active frequencies in the max formulation) disappeared under the epigraph formulation. Second — and absent from the bundle literature, whose iterations are inherently sequential (one trial point, one oracle call, one τ update; for `systune`, one dense Hamiltonian norm per step) — the $|\Omega| \approx 90$–170 constraint evaluations per inner iteration are embarrassingly parallel: the measured speed-up of the oracle is $13\times$ at 32 workers, and at cavity scale a bundle would serialise exactly the expensive part. Parallelism over the exchange set is, to our knowledge, a contribution of this work. What the bundle literature contributed back is recorded in §4: the proximity-control idea reappears as the self-calibrating step governor of §4.4, and [11]'s Nyquist certificate as the winding count of §4.3. Conversely, the campaign's terminal finding (§7, §8) is that the *residual* weakness of the frozen pipeline is exactly where the bundle's convergence theory would have had purchase: the inner solver, still GRANSO run off-label on the (now nearly smooth) restricted problems, stalls at singular-value crossing kinks and on curved constraint boundaries, and the exchange's termination test cannot distinguish "inner solved" from "inner gave up".

3 Problem formulation (summary)

Unchanged from v6, to which the reader is referred for the discretisation, the corrected energy-resolvent definition, the tridiagonal compensator parametrisation, the Sobolev field regularisation, and the analytic gradients. We recall only the objects needed below. The plant is $M\dot{q} = Aq + bu + B_1 d$, $y = c^H q$, $z = q$ (energy norm), with M the SPD mass matrix and A the sparse GL operator at $\mu_0 = 0.96\,\mu_c$ ($\alpha_{\mathrm{ol}} = -0.0177$, $\gamma_{\mathrm{ol}} = 843.2$ certified). The SISO compensator $k(s) = K + n^\top (sI - A_c)^{-1} B_c$ closes the loop through the rank-2 coupling (b, c); the closed-loop pencil is sparse-plus-low-rank, which every tool below exploits. The design vector packs, per case, the compensator row (d, e, f), the output vector n, the gain K, the positions x_a, x_s, and/or the fields ψ_a, ψ_s (NZ = 2–1659 real dofs). Certified lower bounds (v6 §4.3) at $n_e = 200$: *sensor-limited* $\gamma \ge 9.831865$ when the sensor is the default Gaussian, *actuator-limited* $\gamma \ge 9.660005$ when the actuator is, and the absolute two-channel floor $\gamma \ge \sup_\omega \sigma_2 = 3.509066$ for any rank-one control.

4 Method: the frozen pipeline

4.1 Epigraph exchange with a parallel certified oracle

The optimisation is a semi-infinite exchange: minimise t subject to $\sigma_1(\omega_i;\theta) \le t\,\tau_0$ for $\omega_i \in \Omega$, plus the pole cap and stability surrogate rows; solve the restricted problem with GRANSO's BFGS-SQP (constraints explicit, hence generically smooth); certify the candidate over the full band; exchange in the located peaks; repeat until no new active frequency appears and the inner makes no progress. The per-frequency oracle (sparse LU + warm-started weighted power iteration for the full-field channel, Newton polish at peaks) is evaluated in parallel over Ω (96 workers in production; 13× measured at 32). Peak certification uses [6]-style adaptive interval elimination; every reported γ in this document is certified at relative tolerance 10^{-6} over the band $[-30, 30]$.

4.2 The layered certification stack

The single most important structural change since v6 is that certification is now *layered rather than conditional*:

1. **Peak certificate** (per design): γ certified over the band at rtol $= 10^{-6}$ (`certify_results`); refinements within a prior, coarser tolerance are now reported as such rather than as "missed peaks".
2. **Line certificate** (per accepted iterate): the SMV sweep proves no closed-loop eigenvalue *on* $\mathrm{Re}\, s = -\varepsilon$; near-line eigenvalues are harvested into a watch list.
3. **Continuation gate** (per accepted iterate): tracks every watched eigenvalue from the previous accepted design to the candidate by adaptive homotopy with offset-ring shifts and midpoint validation, converting the line certificate into a stability statement *conditional on the chain* — the mechanism that closed the v6 §5.1 blind spot, through which eight unstable designs had been saved.
4. **Winding certificate** (per saved design, §4.3): an *unconditional*, matrix-free count of eigenvalues right of a vertical line — no anchor, no small-steps premise.
5. **Dense cross-check** (GL scale only): full generalized eigensolve, used as ground truth for layers 2–4 and as a save-time veto. It does not exist at cavity scale; its role there is inherited by layer 4.

Layer 5 caught a bug in layer 4 during development (§7, P2) — the stack validating its own newest member is precisely the discipline v6 §5.9 argued for.

4.3 The unconditional winding certificate

Because the closed loop differs from the *decoupled* pair (plant, compensator) by a rank-2 coupling, the characteristic determinant factorises exactly (verified to 10^{-15} against dense determinants):

$$\det(sM_{\mathrm{cl}} - A_{\mathrm{cl}}) \;=\; \det(sM - A)\,\det(sI - A_c)\,f(s), \qquad f(s) = 1 - k(s)\,G(s), \quad G(s) = c^H (sM - A)^{-1} b. \tag{1}$$

The argument principle applied to the region $\{\mathrm{Re}\, s > \ell\}$ then gives an *exact* count,

$$\#\{\lambda_{\mathrm{cl}} : \mathrm{Re}\,\lambda_{\mathrm{cl}} > \ell\} \;=\; W[f; \mathrm{Re}\, s = \ell] \;+\; P_{\mathrm{plant}} \;+\; P_{\mathrm{ctrl}}, \tag{2}$$

where W is the winding number of f along the (closed) contour, P_{plant} is the design-*independent* number of plant eigenvalues right of the line (computed once; at cavity scale it is the physically known handful of near-axis modes), and P_{ctrl} counts $\mathrm{eig}(A_c)$ right of the line (exact, $r \times r$). Each sample of f costs one sparse LU; f' costs one extra solve on the same factorisation. The production line is $\ell = -\varepsilon/2$, so margin-tight designs keep f bounded away from zero. Adaptive phase tracking is hardened by three mechanisms, each added after a measured failure: (i) *mandatory resolution near known poles of* f

(near-line plant modes and A_c poles), where the phase swings by $\sim 2\pi$ over a width equal to the pole's distance to the line; (ii) *midpoint validation* of every accepted step, so an aliased feature must fool three collinear samples; (iii) a *mandatory knot grid* (121 points over the core band plus every hot-pole window edge) that the march may never step past — without which coverage of the central region was *incidental*: the tail truncation W_T scales with $\nu = \|M^{-1}A\| \approx 353$ at $n_e = 200$, single adaptive steps could span hundreds of frequency units, and eight deeply stable designs read winding 0 instead of -1 in 49 points while their neighbours read correctly in 250+. The tail is closed analytically once the certified bound $|G(s)| \le \|c\|\,\|b\|/\big(\lambda_{\min}(M)(|s|-\nu)\big)$ keeps $|kG| < \frac{1}{2}$. After hardening, the certificate reproduces the dense eigenvalue count **exactly on** $104/104$ **designs** across $n_e \in \{50, 100, 200\}$, stable through deeply unstable, coupled and static-gain, at $\approx$ 500–900 sparse solves per design ($\lesssim 1$ s at GL scale). All $27+3$ reported designs are winding-certified with count 0.

4.4 Step governance: the τ governor and the trust-region question

v6 §8.3 measured the inner trust region to be harmful and disabled it — and the v6 §5.1 escapes then showed that *step control had been load-bearing for safety*. The resolution adopted here separates the two roles. Safety-relevant step control now acts on the *accepted-step* level as a bundle-style proximity parameter ([12]): a cap on $\|\Delta z\|$ that is *off until the continuation gate produces evidence* (a homotopy split sets it to $\|\Delta z\|/2$, a gate rejection to $\|\Delta z\|/4$), relaxes after consecutive clean acceptances, and never throttles the inner solver. The inner trust radius, meanwhile, had quietly survived on the warm-start path (`trust_warm`$= 0.1$ by default) as safety-by-stealth; an A/B on the identical warm-started polish of KGC+C measured

	oracle calls	wall time	final γ
trust radius 0.1	33 022	12 639 s	11.7256
trust radius 0	1 666	1 069 s	**11.7180**

— twenty times fewer calls *and* a better optimum. The default is now 0; the gate plus governor carry the safety role explicitly. Two robustness features were added after a four-hour job was lost to a queue time limit: every accepted iterate writes a lightweight checkpoint (verified state only), and start-up auto-resumes from a checkpoint that beats the chosen start, through the standard anchor verification.

4.5 Gauge fixing of complex fields

For cases with a free complex gain the global phase of each field is gauge:

$$(\psi_a, \psi_s, K, n) \;\sim\; \big(e^{i\theta_a}\psi_a,\; e^{i\theta_s}\psi_s,\; Ke^{i(\theta_s-\theta_a)},\; n\,e^{i(\theta_s-\theta_a)}\big) \tag{3}$$

leaves the closed loop identical. Reported and plotted fields are gauge-fixed onto the literal mirror branch (§6) with the residual common phase chosen to maximise $\|\mathrm{Re}\,\psi_a\|^2$; the rotation is applied only after verifying the dense abscissa is unchanged to 10^{-10}. For cases *without* a free gain the field phase is physical and is left untouched.

5 Results

5.1 The frozen table

Table 1 lists all 27 cases: certified γ (rtol $= 10^{-6}$, band $[-30, 30]$), dense-verified margin α, and the optimality classification of §5.2. All designs are dense-verified stable, winding-certified (count 0 right of $\mathrm{Re}\,s = -\varepsilon/2$), band-safe (true peak inside the band each run used), and the full nesting order holds. Headline reductions from $\gamma_{\mathrm{ol}} = 843.2$: factor 86 for gain-plus-positions, factor 215 for gain-plus-fields, the latter within 11.9% of the two-channel floor $\sup_\omega \sigma_2 = 3.509$.

case	γ (certified)	α (dense)	classification
K+FB+FC	3.9255	-0.2164	equalised, 9 active; *mirror-symmetric to* 2.5×10^{-12}
KGC+FB+FC	3.9258	-0.2164	equalised, 7 active; mirror-symmetric
KSP+FB+FC	4.1402	-0.4240	equalised, 8 active; asymmetric (6.7%, $D_c = 0$)
FB+FC	4.3412	-0.2182	equalised, 10 active; *asymmetric member of a mirror pair* (§6)
K+B+C	7.9641	-0.0720	2-active tie (0.18%) with $\lvert\nabla\gamma\rvert \sim 10^4$: *inner-limited, best known*
KSP+B+C	8.2465	-0.0865	3-active tie (0.01%), $\lvert\nabla\gamma\rvert$ elevated: best known
FC	9.6601	-0.2210	at actuator bound 9.660005
KGC+FC	9.6601	-0.2164	at actuator bound
K+FC	9.6602	-0.0873	at actuator bound
K+C	9.6602	-0.0512	at actuator bound
KSP+C	9.6602	-0.0512	at actuator bound
KSP+FC	9.6603	-0.0865	at actuator bound
B+C	9.6745	-0.2031	equalised, 2 active ($\omega = -0.53, -1.83$)
FB	9.8319	-0.2120	at sensor bound 9.831865
KGC+FB	9.8319	-0.2164	at sensor bound
K, KSP	9.8319	-0.058	at sensor bound
K+B, KSP+B	9.8319	-0.056	at sensor bound
K+FB	9.8319	-0.1041	at sensor bound
KSP+FB	9.8321	-0.1036	at sensor bound
C	9.9180	-0.2263	equalised, 2 active ($\omega = +7.99, -0.53$: resonance tie, §5.3)
B	9.9441	-0.2257	equalised, 2 active ($\omega = -0.53, +7.99$)
KGC+B+C	11.4116	-0.1624	single active, $\lvert\nabla\gamma\rvert = 0.45$: inner-limited, best known
KGC+B	11.6009	-0.1519	single active, inner-limited, best known
KGC+C	11.7191	-0.1452	single active, inner-limited, best known
KGC	12.2443	-0.1243	equalised, 3 active

Table 1: The frozen 27-case table ($n_e = 200$, $\varepsilon = 0.05$, band $[-30, 30]$). Every design is dense-verified, winding-certified, peak-certified at rtol $= 10^{-6}$, and band-safe; the nesting partial order holds throughout. "At bound" means agreement with the certified v6 §4.3 lower bound to 10^{-5}–10^{-6} relative.

5.2 Optimality classification, and what it took to earn it

Every row of Table 1 carries one of four verdicts, each backed by a *pair* of independent signatures (a discipline forced by the failures of §7):

- **At a fundamental bound** (14 rows): γ agrees with a certified lower bound of v6 §4.3 to 10^{-5}–10^{-6} relative — global optimality independent of any gradient information.
- **Equalised** (8 rows): several active peaks tied to $\leq 10^{-4}$ *and* a first-order gradient of the magnitude observed at converged equalisations ($O(10)$ in the relative measure). The FB+FC family equalises 7–10 peaks; B, C, B+C and KGC equalise 2–3.
- **Best known, inner-limited** (5 rows): the exchange terminated because the frequency set went stationary while the *inner* solver stopped progressing. For K+B+C and KSP+B+C the design sits at a near-exact multi-peak tie with a relative gradient 10^3–10^4 — three orders above converged equalisations — the measured signature of the singular-value–crossing kink at which BFGS-type inners stall (§8). For KGC+B/C/B+C a single peak is active with a moderate gradient that repeated restarts do not reduce.
- **Symmetric / asymmetric**: the field-case labels of §6.

5.3 What the full-band diagnostics revealed

Until late in the campaign, all curve-based diagnostics ran on gain curves stored over $[-4, 4]$ — a hand-typed constant — while the *certification* always covered $[-30, 30]$. Deriving the diagnostic band

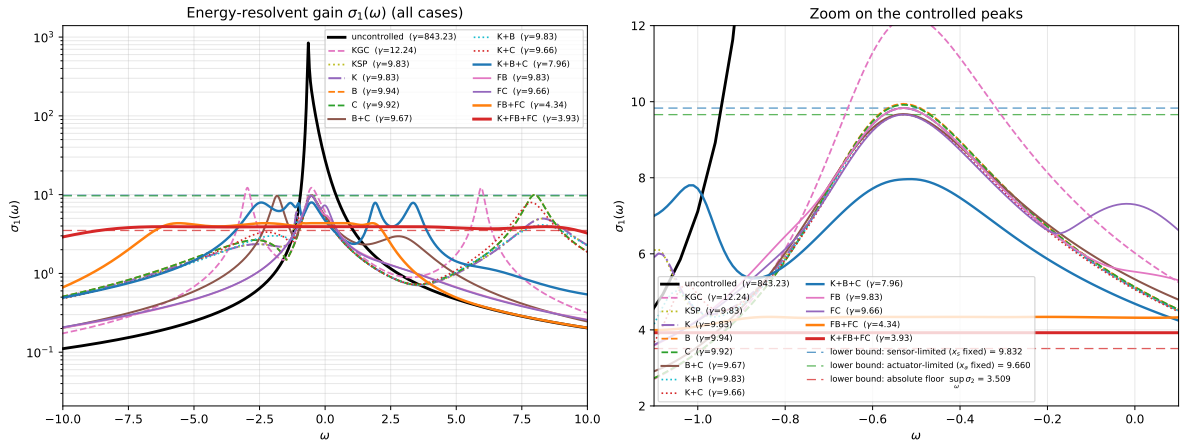


Figure 1: Energy-resolvent gain $\sigma_1(\omega)$ for all cases, computed over the full certified band and plotted on the physically active range (left) with a zoom on the controlled peaks (right). Curves are stored on a nonuniform grid ($\Delta\omega = 0.01$ in the core $|\omega| \le 8$, 0.05 in the wings): the certified band $[-30, 30]$ covers the compensator-pole resonances near $|\omega| \approx 19$, whose role is discussed in §5.3.

from the run records (and storing curves over the run's own band) had an unexpected scientific payoff: the cases B and C, whose "single active peak but gradient not small" verdict had resisted explanation for days, are in fact *exactly equalised* between the main peak at $\omega = -0.53$ and a compensator-resonance peak at $\omega = +7.99$ that the narrow curves could not see; K and KSP show the sibling resonance at $+8.30$ sub-active. The position-only optimiser had, correctly, pushed the main peak down until the resonance became co-active — a two-peak optimum invisible to a diagnostic looking at half the band. The general lesson is recorded as P4 in §7.

5.4 The K+B+C descent: from "stability-limited" to a measured inner limitation

The gain-plus-positions case K+B+C is the row on which most of the campaign's late progress — and three retracted claims — concentrated. Its trajectory, every step triggered by distrust of a figure or a verdict:

γ	α	state	evidence
8.0341	-0.050015	pinned on the stability boundary	$\alpha = -\varepsilon$ to $3 \times 10^{-4}\varepsilon$
		not KKT-stationary	FD-validated $\nabla\gamma \perp \nabla\alpha$, residual 99.3%
		single pinned eigenvalue	vertex hypothesis tested and rejected
		boundary curvature $\approx 8 \times 10^3$	probe drift fits ct^2; straight tangent feasible only $t \lesssim 10^{-5}$
8.0337	-0.050023	restored projected-gradient step	tangent step + Newton restoration onto $\alpha = -\varepsilon$
7.9772	-0.0711	slide: inner resumed from the nudge	-0.71%, design leaves the boundary
7.9641	-0.0720	fresh restarts, then terminal stall	2-active tie 0.18%, $\|\nabla\gamma\| \sim 1.5 \times 10^4$, six peaks within 1.5%

The KKT test (finite differences in the optimiser's own packed coordinates, validated against the analytic gradient chain to 10^{-9}) first *refuted* the constrained-optimum reading of the pinned margin; the projected-gradient probe with restoration then converted the refutation into 0.87% of certified improvement; and the terminal state — interior, multi-peak, enormous gradient, restart-resistant — is the sharpest available measurement of the inner-solver weakness that §8 is tasked with removing. The same mechanism, at one order less severity, describes KSP+B+C (8.2612 → 8.2465).

Workaround versus fix. We emphasise the division of labour in the frozen pipeline. The boundary stall is *detected and certified* by `check_kkt.py` (α-activity plus a large KKT residual against the

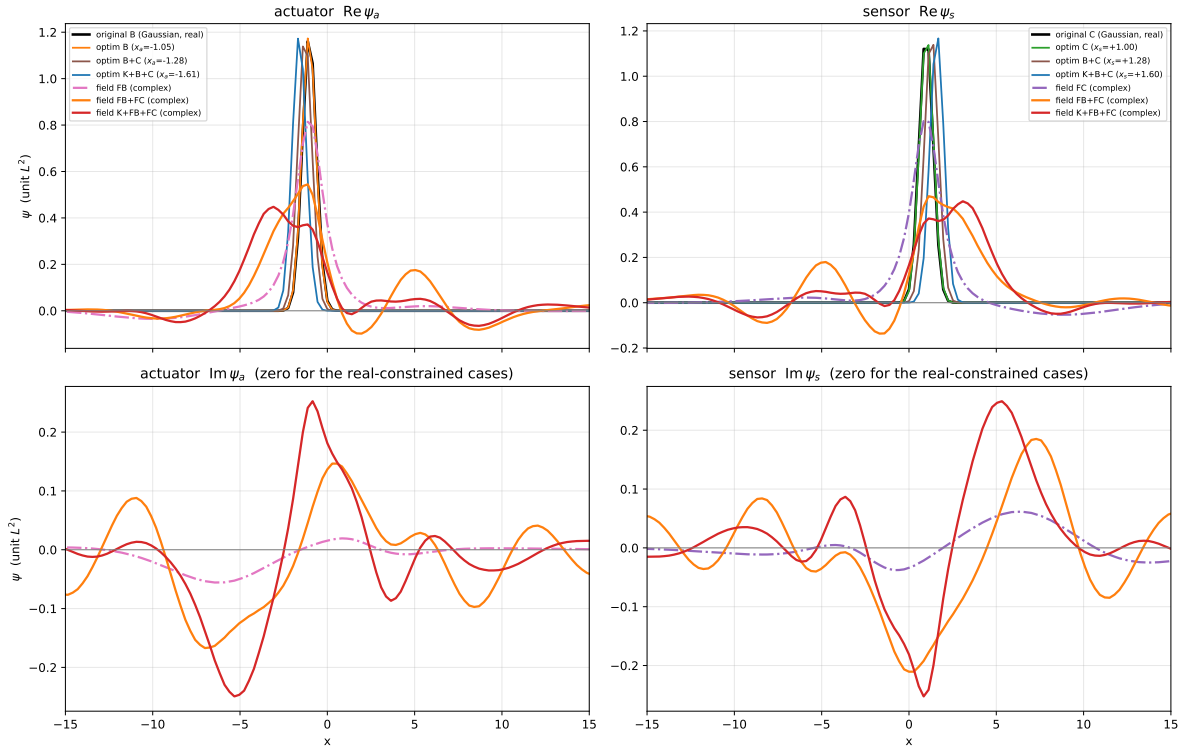


Figure 2: Optimised actuator and sensor distributions (positions and gauge-fixed fields). Complex fields are defined up to a global phase absorbed by the gain; all free-gain fields are plotted on the mirror-symmetric branch with maximal real part (§4.5). The FB+FC curves are genuinely asymmetric (§6).

FD-validated gradients), and *escaped* by a documented manual procedure: a projected-gradient step tangent to the boundary, Newton-restored onto $\alpha = -\varepsilon$, saved as a resume file from which the standard optimiser continues. The optimiser itself was deliberately *not* modified at freeze — stability inside it is still enforced by rejection (gate) plus surrogate rows, so any future run that presses into the stability boundary, at GL or cavity scale, will stall there again until the explicit-constraint inner of §8.1 (item 1) is implemented. The manual loop is the workaround; the constrained inner is the fix; and the choice not to modify the optimiser at freeze time was made so that the 27 validated cases would not have to be re-validated against a changed algorithm.

6 Mirror symmetry: exact restoration and weak spontaneous breaking

This section is new in v7. It began as a question about a figure — the optimised fields *looked* asymmetric — and ended as the campaign's one piece of unplanned physics.

6.1 The symmetry of the problem

Let R denote spatial reflection ($x \mapsto -x$) on the FE grid. The GL operator at this operating point satisfies

$$RA^{\top}R = A \qquad \text{(equivalently } A^{H} = R\overline{A}R\text{)}, \tag{4}$$

verified to 2×10^{-16} in the discrete operator, while M commutes with R. Because the performance channel is full-field and the compensator transfer function is scalar (hence transposition-invariant), the co-design problem is exactly invariant under the antilinear involution

$$S: \quad \psi_a \mapsto R\overline{\psi_s}, \qquad \psi_s \mapsto R\overline{\psi_a}, \qquad x_a \mapsto -x_s, \qquad x_s \mapsto -x_a, \tag{5}$$

with the compensator mapped to its (equivalent) transpose realisation: both γ and α are invariant, verified to 10^{-14} on stored designs. Fixed points of S satisfy $\psi_s = \pm\overline{R\,\psi_a}$ (two branches) and $x_s = -x_a$; for cases with a free complex gain the branch condition holds only up to the gauge of §4.5 (a "phase branch"), and the branch sign/phase is itself gauge. A symmetric-subspace mode of the optimiser parametrises ψ_a alone, ties ψ_s to the chosen branch, and pulls gradients back through the antilinear map (verified against finite differences to 10^{-10}); it halves the field dimension.

6.2 Can the symmetry be broken at a local optimum? Measured answers

Discrete symmetry of the *problem* does not force symmetry of its *optima*: an S-related pair of asymmetric optima is a live possibility, and deciding between the two scenarios was the point of the constrained runs. The campaign's protocol — alternate ("ping-pong") rounds of subspace-constrained and unconstrained polishing, each side resumed from the best available design, iterated until a full round moved either side by less than the certification-derived threshold, with all endpoints certified at rtol $= 10^{-6}$ on a common band — produced a clean split:

Free complex gain: the optima are exactly symmetric. For K+FB+FC and KGC+FB+FC the unconstrained optimiser, released from a symmetric start, *stays on the subspace*: the final full-space optima satisfy $\|\psi_s - \overline{R\psi_a}\|/\|\psi_s\| = 2.5 \times 10^{-12}$ (Fig. 3, first two columns), and the constrained search reaches the same γ faster at half the field dimension. The apparent asymmetry that started the investigation was *optimiser slack in a flat direction* (a 39% field asymmetry was worth 1.4% in γ at an intermediate iterate) plus *gauge*: on the phase branch, Re/Im panels of an exactly symmetric design look wildly asymmetric, which is why reported fields are gauge-fixed (§4.5).

Fixed compensator: weak spontaneous symmetry breaking. For FB+FC the fixed compensator leaves no gauge — the field phase is physical — and the converged landscape is genuinely asymmetric:

- a **mirror pair** of asymmetric optima at $\gamma = 4.3412$ (the stored design and its S-image, exactly degenerate by the verified invariance), field asymmetry 39%, *exactly equalised* over 10 active peaks;
- a **distinct symmetric local optimum** at $\gamma = 4.3417$, i.e. 1.1×10^{-4} relative above the pair, itself exactly equalised over 9 active peaks and *full-space* first-order stationary;
- a **separate basin**: unconstrained descent started *at* the symmetric optimum returns to it (12 outers, no improvement), so the symmetric point is not a saddle on a path to the pair;
- the direct midpoint of the mirror pair evaluates at $\gamma = 4.4016$: the pair is far apart in design space, with the symmetric valley a distinct, nearly-as-good basin.

The breaking is *weak* — depth 1.1×10^{-4} relative, resolved at roughly $100\times$ the 10^{-6} certification tolerance, though endpoint re-certification drifts of up to 6×10^{-5} were observed, so the claim rests deliberately on three independent legs (certified gap + basin separation + two-sided exact-equalisation stationarity) rather than on the gap alone. The strictly proper case KSP+FB+FC ($D_c = 0$) sits in between: residual asymmetry 6.7% that gauge alignment cannot remove, at a γ penalty below 10^{-3}; it was not pushed through the constrained protocol.

Interpretation. Enlarging the controller class by a single free complex gain *restores* the symmetry exactly: the gauge orbit connects would-be mirror pairs, and the optimiser lands on the symmetric representative to 10^{-12}. Removing that freedom (fixed compensator) lets the optimum break the mirror symmetry spontaneously, by a measured 10^{-4}. Practically, the symmetric parametrisation is the recommended production mode for symmetric plants: same or better optimum, half the field dofs, faster convergence — with the caveat, now measured, that the assumption must be *checked* per controller class rather than assumed.

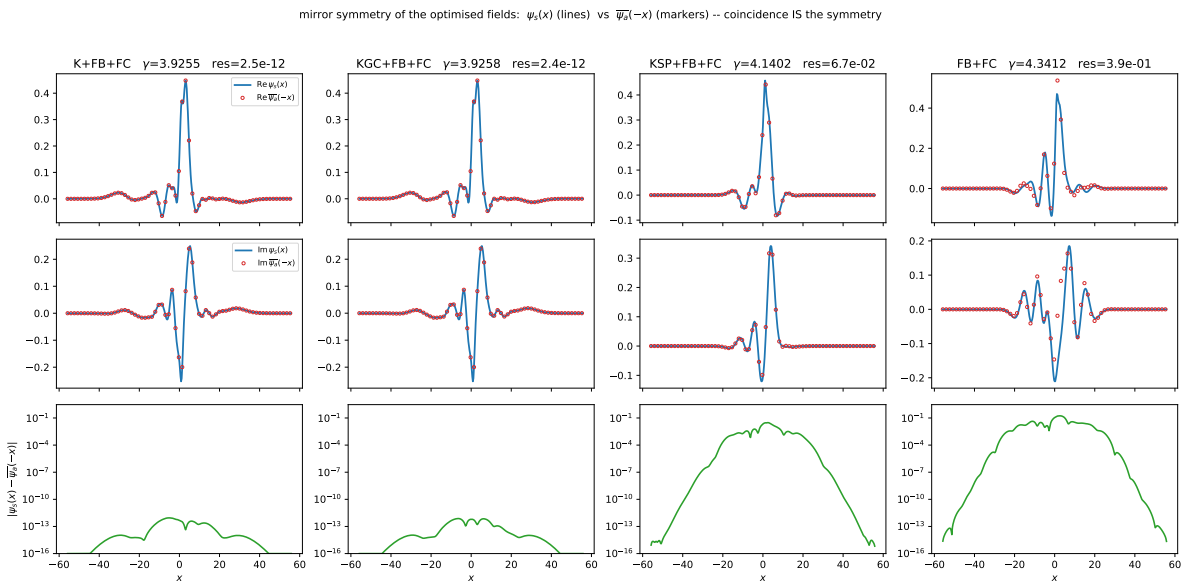


Figure 3: The symmetry evidence figure: $\psi_s(x)$ (lines) overlaid on $\overline{\psi_a}(-x)$ (markers), with the pointwise residual below (log scale). Coincidence *is* the symmetry: K+FB+FC and KGC+FB+FC at residual 2.5×10^{-12}; KSP+FB+FC (6.7%) and FB+FC (39%) genuinely asymmetric.

7 Pitfalls, and how they were caught

v6's §5 ("revisions forced by validation failures") is continued here. Each entry records symptom, diagnosis, fix, and the transferable lesson; the meta-lesson is stated at the end.

P1 — Conditional stability certification (v6 §5.1, closed for good). The line sweep proves the line clear, not the half-plane empty; after the trust-region ablation the small-steps premise of the continuation argument silently failed and eight unstable designs (dense α up to $+4.3 \times 10^6$) were saved as results. Fixed in stages: continuation gate with adaptive homotopy and anchor verification; dense save veto at GL scale; and finally the *unconditional* winding certificate of §4.3, which removes the last conditional link and ports to the cavity. *Lesson: a certificate must be unconditional, or every premise it rests on must itself be guarded.*

P2 — The winding tracker's incidental coverage. The new certificate initially contradicted dense on eight deeply stable designs (winding 0 instead of -1 in 49 points). Diagnosis: adaptive stepping bounded only by the tail truncation could leap the entire active band; whether the march slowed in the centre depended on whether a sample happened to *land* near the structure. Fix: the mandatory knot grid — coverage made deterministic. The bug was caught by the dense cross-check within the first full-suite run, i.e. by the layer the certificate will replace at cavity scale. *Lesson: validate every new certificate against ground truth at benchmark scale, in the regime where ground truth still exists.*

P3 — Safety-by-stealth: the warm-start trust radius. The trust region, ablated as harmful, had survived on the warm-start path and throttled every polish and every symmetric run (a 10^{-4}-per-outer creep in one four-hour log). The A/B of §4.4 measured 20× fewer calls at better γ; the default is now off, with the safety role carried by explicit, evidenced mechanisms. *Lesson: when a mechanism is removed for performance, list every role it was quietly playing and reassign each explicitly.*

P4 — Hand-typed constants drifting from the source of truth. Four instances of one genus: diagnostic gain curves stored on a hand-typed $[-4,4]$ while certification covered $[-30,30]$ (hid the resonance equalisation of §5.3 and mislabelled B/C for days); a hardcoded controller order $r = 7$ against the seed catalogue's $r = 8$ (harmless until the first code path that used it); the plot band and the certification band maintained separately; and a cache-refresh trigger that tested coverage but not resolution, so the first fix (wide but coarse curves) was itself certified as fine by the second bug. All four are now derived from the run records or the design files. *Lesson: derive from where it is decided; and a cache check must test every property the cache is supposed to guarantee, not the one that failed last time.*

P5 — Verdicts asserted from single signatures. Three diagnostic claims were, in sequence, refuted by their own follow-up tests: "stability-limited: the gradient is the multiplier" (refuted by the FD KKT test — gradients orthogonal, not anti-parallel); the single-constraint KKT geometry itself (the probe left the "tangent" quadratically: not a multi-eigenvalue vertex, as first suspected and tested, but a boundary of curvature 8×10^3); and "equalised: classic H_∞ optimum" for K+B+C (a 2-active tie with a gradient $10^3\times$ converged cases is a stall's way-station, not an optimum). Each refutation was converted into either certified improvement (0.87% on K+B+C) or an honest label. Every verdict in Table 1 now requires two independent signatures (§5.2); a scope-leak bug in the verdict code, which briefly stamped every row with the same bound, was itself caught because its output was absurd. *Lesson: a diagnosis is a hypothesis; the tool that asserts it must also ship the test that could refute it.*

P6 — Lost work and silent termination. Long jobs saved results only at the end (one four-hour improvement lost to a time limit); fixed by per-acceptance checkpoints with verified-state semantics and auto-resume. Separately — and unresolved by design, see §8 — the exchange's termination test cannot distinguish "inner solved the restricted problem" from "inner gave up on it": the run log shows the inner's claimed objective *below* the certified value at the terminal iterates of the stalled cases, i.e. the inner under-reports its constraint violation, the design stops moving, no new frequencies appear, and stagnation reads as convergence. The termination message now says so (`TERMINATED: inner solver at its limit -> best known` rather than `CONVERGED`); fresh restarts recover $\sim 1.6 \times 10^{-3}$ per round before saturating.

The meta-lesson. Nearly every pitfall above was surfaced by a human distrusting an artefact — an asymmetric field panel, an under-sampled curve, peaks that "should" be equal, a verdict that proved too much — and every act of distrust either fixed a tool or improved a certified number. The pipeline's job is to make such distrust cheap: every figure is regenerable from the frozen archive, every verdict names its evidence, and every claim in this report is attached to a command that re-derives it.

8 Perspectives: what must be implemented next

8.1 The inner solver is now the binding constraint

The frozen pipeline's residual weakness is localised and measured. GRANSO's BFGS-SQP is run off-label: designed for nonsmooth max formulations with a steering penalty and, by its authors' statement, no convergence guarantees [13], it here faces nearly smooth restricted problems — and fails at exactly the two places smoothness breaks:

- *Singular-value crossing kinks.* At multi-peak ties the active-set description changes discontinuously; K+B+C terminates at a 0.18% tie with a relative gradient of 1.5×10^4, six peaks within 1.5%, and 64 free parameters — a configuration from which restart-with-fresh-state descends a further $\sim 1.6 \times 10^{-3}$ per round before saturating, demonstrating that the landscape has more to give than the inner extracts.

- *Curved constraint boundaries.* Stability is enforced by rejection (the gate) plus surrogate rows; a rejection mechanism cannot slide *along* the boundary. The K+B+C episode measured the consequence: a first-order descent direction of relative slope 0.5 tangent to a boundary of curvature 8×10^3, exploitable only through manual projected-gradient steps with Newton restoration.

The replacement, planned since the review of §2 and now specified by measurement, is a smooth constrained NLP inner:

1. **Explicit constraints**: $\sigma_j(\omega_i;\theta) \le t\tau_0$ over the exchange set, the pole cap, *and the stability margin* $\alpha \le -\varepsilon$ *as a smooth constraint* (rightmost-eigenvalue value and gradient are available from the machinery of the KKT tool), so that boundary-following is native SQP behaviour rather than a manual probe.
2. **Second-singular-value enrollment**: when the gap $\sigma_1 - \sigma_2$ at an active frequency falls below a threshold, enroll $\sigma_2(\omega_i) \le t\tau_0$ as a second constraint — each σ_j is then smooth again and the kink is represented in the active set instead of hidden in a max. Several frozen optima show gaps of 10^{-2}–10^{-1}; this is where a plain SQP would otherwise chatter.
3. **Solver**: IPOPT via `cyipopt` with a limited-memory Hessian — the dense BFGS of GRANSO is $O(\mathrm{NZ}^2)$ and will not survive the cavity's $\mathrm{NZ} \sim 10^4$–10^5 field dofs — with the exchange, the parallel oracle, and the entire certification stack unchanged around it.
4. **Remez–Newton endgame**: at a near-equalised configuration, solve the equalisation system $\{\sigma_1(\omega_i) = t\tau_0 \text{ (active)},\ \text{stationarity}\}$ by Newton for quadratic terminal convergence; all ingredients (peak frequencies, gradients, multipliers) are already computed by the diagnostics.
5. **Cheap mitigation, immediately**: automatic inner retry with fresh state when the exchange is about to terminate on an inner stall — the internalised version of the restart rounds that recovered the K+B+C improvements.

Acceptance test, fixed in advance: solve K+B+C to an equalised optimum with a relative gradient of $O(10)$, without manual restarts, matching or beating $\gamma = 7.9641$; and reproduce the remaining 26 certified values of Table 1 within 10^{-4}. The frozen suite is the regression benchmark.

8.2 Cavity-scale hardening of the certificates

- *Winding certificate at scale*: P_{plant} and the hot-pole list are design-independent inputs (the physically known near-axis modes); the tail constant $\nu = \|M^{-1}A\|$ grows like h^{-2} under refinement, so the tail bound should move to the M-weighted norm, and the per-design cost ($\sim 10^3$ sparse solves) parallelises over line segments.
- *Disk sweeps*: the adaptive elimination is inherently sequential (measured $1.43\times$ at 8 workers); speculative evaluation of k shifts spread over the uncovered intervals — every returned disk is valid regardless — trades redundant solves for scalability, in the spirit of [5]'s parallel level sets.
- *Oracle acceleration*: a reduced-basis / greedy-interpolatory model of the parametrised resolvent [4] across exchange iterations, with certification kept full-order; the warm-started singular-vector reuse already in the oracle is the degenerate version.

8.3 Beyond a single operating point

The `systune` progress-function pattern $\Phi_\eta = \max\{f, \eta g\}$ [9] is the natural cast for the cavity's multiple requirements (margins, bandwidth caps), and the multi-model parametric-robust extension [10] for robustness over *Re*; both compose with the exchange as additional max terms and inherit its parallelism. Finally, the symmetric-subspace mode of §6 should be the default for symmetric plants — with the per-class symmetry check of §6 run once, since this campaign demonstrated that the answer can genuinely differ between controller classes.

9 Reproducibility

The frozen archive (`frozen_20260826.tar.gz`) contains all designs (`results/`, `results_sym/`), logs, figures, and the complete toolchain: optimisation (`run_case_epi.py`, `packer.py`, `parallel.py`), certification (`certify_results.py`, `certify_sparse.py`, `winding_cert.py`, `verify_dense.py`), diagnostics (`check_optimality.py`, `check_kkt.py`, `check_band.py`, `check_symmetry.py`, `collect.py`), symmetry and presentation (`gauge_fields.py`, `plot_symmetry.py`, `plot_results.py`), and the harvest recipe (`sweep.sh`). Every number in this report is re-derivable by the chain `verify_dense` → `winding_cert` → `certify_results -rtol 1e-6` → `collect` → `check_optimality -gbounds`; every figure by `plot_results`/`plot_symmetry` on the archived data.

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